

22/12/2025

9a) WAP to traverse a graph using BFS method.

```
#include <stdio.h>
```

```
int graph[20][20], visited[20], n;
```

```
void BFS (int start) {
```

```
    int queue[20], front = 0, rear = 0;
```

```
    visited[start] = 1;
```

```
    queue[rear++] = start;
```

```
    while (front < rear) {
```

```
        int node = queue[front++];
```

```
        printf("%d ", node);
```

```
        for (int i = 0; i < n; i++) {
```

```
            if (graph[node][i] == 1 && !visited[i]) {
```

```
                visited[i] = 1;
```

```
                queue[rear++] = i;
```

```
            }
```

```
        }
```

```
int main () {
```

```
    int start;
```

```
    printf("Enter no. of vertices:");
```

```
    scanf("%d", &n);
```

```
    printf("Enter adjacency matrix:\n");
```



```

for (int i=0; i<n; i++){
    for (int j=0; j<n; j++){
        scanf ("%d", &graph[i][j]);
    }
    for (int i=0; i<n; i++){
        visited[i] = 0;
    }
    printf ("Enter starting vertex: ");
    scanf ("%d", &start);
    printf ("BFS Traversal:");
    BFS (start);
    return 0;
}

```

Output:

Enter no. of vertices: 5

Enter adjacency matrix:

0	1	0	1	0
1	0	1	0	1
0	1	0	0	0
1	0	0	0	1
0	1	0	1	0

Enter starting vertex: 0

BFS Traversal: 0 1 3 2 4